

Onomatangiplay : A Game Experience with a Tangible Auditory User Interface Created by Representing Game Effects with a Real-world Sound Source

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Figure 1: Overall game experience

ABSTRACT

Since the advent of augmented reality, video game representations have been extended beyond the display. Many studies have used real objects to assist in the presentation of visual and haptic information, resulting in video games that are integrated with reality. Researchers often use tangibles UIs as a method to extend visual and haptic information presentation, but they commonly use speakers for auditory information presentation, and tangibles UIs dealing with real objects have few applications except for some music applications.

In this study, we developed a device that presents game sound effects using real object sound sources. By assigning a real object sound source to each of the game sound effects, not only can auditory information be presented instead of speakers, but also a tangible UI can be realized where the user can access the video game by touching the sound source. We named this tangible UI as

Tangible-Auditory User Interface (t-AUI), referring to GUI, and CLI. The production of t-AUI will enable the application of tangible UI to sound generating devices such as holding down and blocking the output of video game sound effects with hands, and reproducing sound effects with real objects for video game operation.

CCS CONCEPTS

• **Human-centered computing** → *Sound-based input / output; Mixed / augmented reality.*

KEYWORDS

Game interaction, Tangible user interface, Auditory user interface

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1 INTRODUCTION

Auditory information plays a major role in the video game experience. For example, in FPS games, if you play with headphones, you can guess the location of enemy players by the direction of their

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footsteps. This means that the sound effects are not just a product, but have information that affects the gameplay. However, it is a case of how to utilize the sound effects received by the player, and the sound effects are not the starting point of the game interaction. There are several devices for using sound in-game interaction. An example of a game device that was capable of voice input in the past is the GameCube microphone, a peripheral device for the Nintendo GameCube. The Nintendo DS also had a microphone attached to the main unit, and voice input was possible using that microphone. However, it was only a function for accepting input. As augmented reality began to be applied to video games, video game representations were extended into real space. Multi-modality in video game experiences has been shown to increase user satisfaction[7]. It is also said to be able to make users focus on video games[9]. Even before augmented reality was used in video games, many arcade games used real objects to create play experiences such as whack-a-mole and air hockey. However, both were mostly complementary to visual and haptic information. In terms of auditory information, although research has progressed in areas such as three-dimensional acoustics, the pursuit of expression is based on the use of speakers, and the approach to output has not changed. Ishii et al. have proposed the concept of tangible[6]. In the field of music, there is much research on the representation of tangible sounds. In the field of music concretes, there is an attempt by Kanta Horio to “Design for sound performance devices based on concrete;te sound.” [5]. In this study, we apply his approach to interaction design in video games. In this study, we created an interface that enables a tangible representation of game sound effects. By applying a real object sound source to each of the game sound effects, not only does this provide an alternative output to speakers, but it also allows the user to access the video game by touching the real object sound source, allowing input like a game controller. We named this interface Tangible-Auditory User Interface (t-AUI). The t-AUI enables bidirectional input and output between the video game and the user, enabling “Object Sound” for output and “Manipulation” and “Holding” interaction for input. The three interactions will be explained in detail. In “Object Sound” we have created an alternative to speakers for sound expression. By expressing sound effects with real objects, users can visually understand the connection between “events” and “object sounds”. This will allow the user to determine which sound effects should be accessed in “manipulation” and “Holding”. “Manipulation” serves as a new game controller. Users can influence video games by recreating sound effects using real object sound sources. Conventional voice input in video games allowed only a single input because there was only one microphone for gameplay. t-AUI, however, allows multiple inputs simultaneously because it has a real object for each sound effect. “Holding” is an interaction that is a major feature of t-AUI. Users can mute the sound effects emitted from video games by physically holding down the t-AUI. With speakers, there is only one speaker for every sound, so when you mute a sound effect, the entire sound effect is muted. However, with t-AUI, each sound effect is assigned to a real object, so users can mute sound effects individually, and the situation can be reflected on the related object in the video game.

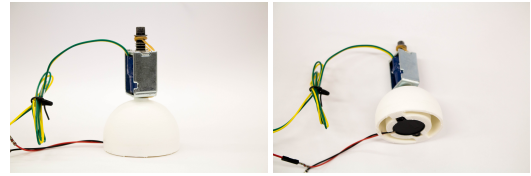


Figure 2: t-AUI footstep module implemented this time

2 RELATED WORKS

There are many previous works of TUIs being used in video games. Jiwoo Hong et al. in “A Demonstration of Conveyor World: Mixed Reality Game on Physically Actuated Game Stage”[4] proposed a scrolling action game in which real objects placed on a conveyor belt serve as obstacles. In “Miniature Haptics: Experiencing Haptic Feedback through Hand-based and Embodied Avatars”[10], Bo-Xiang Wang et al. propose an experience of playing a soccer game by touching conductive aluminum foil balls with hands as avatars. “Makey Makey”[1] by Beginner’s Mind Collective proposes a new platform that allows you to create a gamepad with tangible UI. In “FS-Pad”[8], the player character’s knockback is reproduced by spontaneously moving the thumb stick of gamepad using a motor. “SmartPads”[2] proposes to combine pads to create TUIs of various shapes and operations that can be performed by stepping on them. However, most of the TUIs in video games only extend the presentation of visual and haptic information, and there are few examples of the extension of auditory information presentation. There are many examples of TUI research in the music field. In the field of music concretes, there is an attempt by Kanta Horio to “Design for sound performance devices based on concrete;te sound.” [5], in which he proposes a performance device that makes use of the relationship between physical “events” and “object sounds”. Also, “The Kitsch-Instrument: Hackable Robotic Music”[3] uses solenoids to create a musical expression that is played by tapping everyday objects. In this study, we applied such an approach to interaction design in video games.

3 ONOMATANGIPLAY

3.1 Device

In our project, we created a t-AUI specifically for footsteps made by players in video games (Figs.2). It consists of a solenoid, a vibration sensor, and a cup produced by 3D printing (Figs.3). This work was created regarding the use of bowls to create sound when reproducing footsteps as a sound effect.

3.2 Interaction

Serial communication between Unity and Arduino was used to achieve interaction. The t-AUI realizes “Object Sound” by extruding motion with a solenoid, and obtains information for “Manipulation” and “Holding” by detecting vibration with a vibration sensor. In the “Object Sound” mode, the solenoid connected to the Arduino is activated by the signal from Unity to reproduce the footstep



Figure 4: “Object Sound” interaction. The module make the game sound effects sounds by real objects. The module activated and sounding by itself.



Figure 5: “Manipulation” interaction. By hammering the module against the desk, it effects game play.



Figure 6: “Holding” interaction. By holding the module by hand to prevent it from making a sound, game sound is muted and game play is affected in the game.

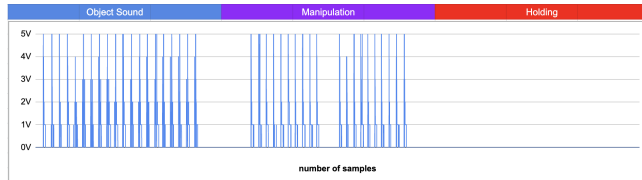


Figure 7: Vibration status for each interaction



Figure 8: A game character running and the module actuate depending on key input.



Figure 9: By hammering the module against the desk, a game character is stamping his feet.



Figure 10: By holding the module by hand to prevent it from making a sound, a game character is moving in stealthy steps.

sound as shown in figure.4. For “Manipulation”, as shown in figure.5, you pick up the module, hit it and make a knocking sound. In the “Holding” mode, as shown in figure.6, the solenoid is suppressed to prevent it from moving, thus blocking the sound. If the vibration sensor reacts when the “Object Sound” signal is not sent, it is judged that “Manipulation” is being performed. If the vibration sensor does not respond even though an “Object Sound” signal is being sent, it is considered to be in a “Holding” state(Fig.7).

We named our t-AUI-based video game experience Onomatangiplay (Fig.1). We created a demo game in Unity and created three types of interactions related to footsteps. Operate using both t-AUI and key input. “Object Sound” is made to match the time of the run (Fig.8). You can step on the spot by doing “Manipulation” without entering the left and right arrow keys (Fig.9). Enter the left and right arrow keys while “Holding” to become stealthy steps (Fig.10).

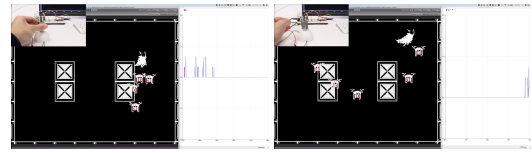


Figure 11: The player is trying to sneak up and catch a target.

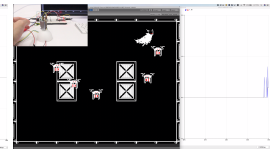


Figure 12: The player is running to catch a target, and the target is running away.

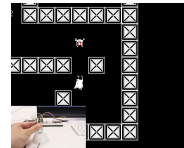


Figure 13: The player is sneaking around to avoid being spotted by the enemy.

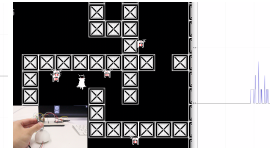


Figure 14: The player diverts enemies by stepping on their feet.

4 USER EXPERIENCE

We implemented game contents incorporating three types of interaction with t-AUI. We developed two games: a game to catch escaping targets, and a game to reach the goal without getting caught by enemies. We named them “Catching target” and “Spy Game”. They are mini-games with a downward view.

In “Catching target”, the target moves away from the player in response to “Object Sound”. By using “Holding”, you can approach and catch the target without being detected (Fig.11). You can move faster by running, but your footsteps will sound and the target will run away (Fig.12). If you catch five of them in time, you complete the game. In “Spy Game”, you have to avoid getting caught by enemies and try to get to the exit, enemies reacts to the sound of footsteps and approaches. If you get caught by enemies on the map, the game is over. It is possible to advance undetected by the enemies by stealthy footsteps with “Holding” (Fig.13), and to create a diversion by footsteps with “Manipulation” (Fig.14). It also allows you to move fast by running, although it makes footsteps sound. It is a content that uses the three interactions to get to the target point.

5 FUTURE WORK AND LIMITATIONS

The only t-AUI we were able to use with the video game was footsteps. To increase the variety of sound effects in the future, we would like to consider ways to make sounds other than by tapping, and ways to detect sounds other than by vibration sensors. In the Japanese Kabuki theater, there are examples of stage direction in terms of acoustics using an equipment called an “ongu”, and we plan to use this as a reference. For example, since there is a sound device called a rain stick that reproduces the sound of rain, a game that uses the sound of rain to control the water level could be considered.

As for the user experience, there is a problem that both hands are occupied because the keys and t-AUIs are operated at the same

time in the current situation. It is considered that the difficulty of the game experience increases when multiple t-AUIs are used. We would also like to conduct user interviews because we need to verify the validity of t-AUI and game interaction.

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